

Katlyn Wang

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EDUCATION

University of California San Diego, San Diego, CA

Expected Graduation: June 2026

Master of Science – Computer Science GPA: Cumulative 3.91/4.00

University of Michigan – Ann Arbor, Ann Arbor, MI

Graduation: May 2024

Bachelor of Science – Computer Science GPA: Cumulative 3.84/4.00, Major 3.9/4.00

- Relevant coursework: Foundations of Large Language Models, Computer Vision, Web System, Intro to Machine Learning, Database Management Systems, Human Centered SW, Data Structures and Algorithms, Applied Regression Analysis

SKILLS

Programming & Web: Python, C++, Java, JavaScript (React, Node.js), REST API, SQL, Firebase, Flask, HTML/CSS, SPL
Cloud & DevOps: AWS (Lambda, S3, AppConfig, DynamoDB), Docker, CI/CD (Git), Agile, Linux (Ubuntu), Splunk
ML & Tools: PyTorch, TensorFlow, transformers (Hugging Face, BERT), NumPy, VS Code, Figma, Jira, Notion, Excel

WORK EXPERIENCE

Software Development Engineer Intern

June 2025 – Sept 2025

Amazon.com, Inc, Global Payments Tech

Seattle, WA

Led the delivery and production deployment of a customer-facing, **self-service** “1-token” testing platform serving for **over 1000** plugin registrations; tests auto-trigger on **AWS** cache updates, eliminating a manual touchpoint, and cutting end-to-end processing time for each **from 12 hours to 15 minutes**

- Designed and developed backend **Java** APIs, **batch operations**, and React UI pages to manage allowlists; used **AWS Lambda** as a serverless integration layer from frontend requests to **S3** events and orchestrate **AppConfig**-driven test workflows.
- Built monitoring and **analytics** for test results, usage patterns, and metrics for data-driven rollouts and reliability improvements.

AI Engineer Intern

July 2024 – Sep 2024

ThingX Technology

Hong Kong

- Contributed to the human fall detection program through real-time thermal monitoring, building and training a **multimodal model** to expand the training dataset by simulating thermal videos from RGB inputs
- Collaborated on modifying the **ControlNet** architecture to support both image and video RGB inputs alongside text descriptions. Leveraged and **fine-tuned** pretrained **multimodal** models, enhancing **performance** and accuracy to successfully generate thermal outputs that accurately align with RGB video inputs.

Software Engineer Intern

May 2023 – Aug 2023

Orange Cyberdefense

Shanghai, China

- Threat Hunted through the Portfolio Brand CRM and web logs database in **Splunk** to build over 20 use cases in 5 routine categories using **Python** and **SPL** for regular detection on potential security vulnerabilities and malicious attacks
- Improved performance of database for internal security system by 32% and rebuild database analysis for future tracking

PROJECTS

Mobile App Developer, *Settlyfe, San Diego, CA*

Dec. 2024 – present

- Co-developed a cross-platform mobile application using **React Native**, building a **full-stack** solution with five core pages and over 20 sub-pages to streamline property and service listings, improve price transparency, and simplify the rental process.
- Planned the groundwork for a **Recommender System** that will help owners, agents, and renters with personalized property suggestions, optimizing user engagement and simplifying the decision-making process.

Landscape image classification with Convolutional Neural Network

Oct. 2023 – Nov. 2023

Developed and enhanced a **CNN**-based image classification system using **PyTorch** to differentiate between photographs of the different landscapes, facilitating an efficient scrapbooking process for a vacation. Executed data preprocessing and leveraged **CNN** visualization to improve efficiency and utilized **transfer learning** and **data augmentation** to boost model precision and resilience.

- Achieved 0.85 in AUROC test score with optimizing model architectures on filter number selection, weight decay, visualizing with Grad-Cam algorithm, and applying transfer learning and data augmentation strategies like rotate and grayscale the image.
- Enhanced performance to 0.94 in AUROC by adding **dropout** layers, **freezing** layers, and **augmenting** images' bright and contrast.

Video Review text classification with SVMs in Natural Language Processing

Sept. 2023

Implemented and optimized text classification models from **SVM** and **LinearSVM** in **sklearn** to classify sentiments of video reviews and ratings, aiding a critical movie enthusiast in efficiently automating the movie selection process. **Utilized feature extraction**, and addressed class **imbalance** and bias through **hyperparameter tuning** and asymmetric cost function analysis to optimize.

- Achieved AUROC and F1 over 0.95 on binary classification using linear/polynomial SVMs with 5-fold CV, hyperparameter tuning, and class-weight adjustments; extended to multi-class with word-count and TF-IDF features yielding AUROC over 0.8